

Documents, Context Rot, and How Models Fail

THE IDEA IN ONE LINE

Accepting a file is not reading it, and models fail in patterns you can learn.

FILE FORMATS, ROUGHLY

FORMAT	WHAT HAPPENS	USE WHEN
Plain text / markdown	Cleanest read; structure preserved if headers are marked	Always, if you can get it
PDF (text layer)	Usually fine for prose; tables, footnotes, and multi-column layouts often scramble	Prose-heavy documents
PDF (scanned image)	Model may see little or nothing and still answer confidently	Avoid; OCR first
.docx / .odt / spreadsheets	Depends on the tool's converter; check a known value	Confirm before trusting

THREE HABITS

- Fresh threads.** New topic, new conversation. Long chats accumulate stale instructions.
- Summarize and restart.** In a long thread: `Write a 200-word summary of where we are and what's decided.` Paste that into a new chat.
- Extract, then reason.** Pull the relevant pages or table into a clean document first. Ask the hard question of the small document.

SIX WAYS IT GOES WRONG

FAILURE	WHAT IT LOOKS LIKE	THE CATCH
Fabrication	Invented citation, quote, number, URL	Verify two citations at random. If one fails, verify all.
Misreading	Table value wrong, footnote missed, scan unread	Ask one question you know the answer to.
Overreach	Answers a question the source doesn't address	What in the source supports this? Quote it.

FAILURE	WHAT IT LOOKS LIKE	THE CATCH
Drift	Long conversation, old instruction still active	Fresh thread. Summarize and restart.
Sycophancy	Agrees with your bad premise	Argue that my premise is wrong.
Silent substitution	Did an easier, adjacent task	Restate the task you completed in one sentence.

THE PRE-FLIGHT

Before trusting any output you'll use:

- Does it answer what I asked, or something nearby?
- Can I point to where each key claim came from?
- Did I check one thing I already knew?
- Is this conversation older than the topic?

PITFALLS

- Uploading twelve documents and asking one question. Retrieval gets worse with each file.
- Assuming the failure was the model's. Often the spec was missing a constraint. Check the spec first.
- Not telling anyone. A failure you hide is a failure the next person repeats. Log it.

GO DEEPER

- Chroma, "Context Rot" (the study): trychroma.com/research/context-rot
- Liu et al., "Lost in the Middle" (the earlier paper): aclanthology.org/2024.tacl-1.9
- Anthropic, "Effective context engineering for AI agents" (why context is a finite resource): anthropic.com/engineering/effective-context-engineering-for-ai-agents
- The group Playbook, Lies Log section.